	Numbers & Shapes	Colors & Numbers	Numbers & Size	Colors & Shapes	Size & Shapes	Colors & Size	Average
Random Baseline	25.0	25.0	25.0	25.0	33.3	25.0	26.4
Qwen-VL-Chat	29.5	26.5	26.5	26.5	30.0	31.0	28.3
LLaVA-13B	20.0	30.0	31.5	36.0	**39.0**	30.0	31.1
Gemini Pro	29.0	21.0	29.5	27.5	36.5	37.0	30.1
Claude 3 Opus	48.0	54.5	**34.0**	42.5	33.0	50.0	43.7
GPT-4V	**52.5**	**56.0**	31.0	**47.0**	31.5	**55.0**	**45.5**

Table 2: Accuracy of large multimodal models for dual-concept abstract patterns in PUZZLEVQA.

multimodal model based on existing benchmarks (Yue et al., 2023). We use their publicly available API to query the “gpt-4-vision-preview” version of the model.

5 Results

We report the main evaluation results on single-concept and dual-concept puzzles in Table 1 and Table 2 respectively. The evaluation results for single-concept puzzles, as shown in Table 1 reveal notable differences in performance among the open-source and closed-source models. GPT-4V stands out with the highest average score of 46.4, demonstrating superior abstract pattern reasoning on single-concept puzzles such as numbers, colors, and size. It particularly excels in the "Numbers" category with a score of 67.5, far surpassing other models, which may be due to its advantage in math reasoning tasks (Yang et al., 2023). Claude 3 Opus follows with an overall average of 39.4, showing its strength in the "Shapes" category with a top score of 44.5. The other models, including Gemini Pro and LLaVA-13B trail behind with averages of 34.5 and 27.5 respectively, performing similarly to the random baseline on several categories.

In the evaluation on dual-concept puzzles, as shown in Table 2, GPT-4V stands out again with the highest average score of 45.5. It performed particularly well in categories such as "Colors & Numbers" and "Colors & Size" with a score of 56.0 and 55.0 respectively. Claude 3 Opus closely follows with an average of 43.7, showing strong performance in "Numbers & Size" with the highest score of 34.0. Interestingly, LLaVA-13B, despite its lower overall average of 31.1, scores the highest in the "Size & Shapes" category at 39.0. Gemini Pro, on the other hand, has a more balanced performance across categories but with a slightly lower overall average of 30.1. Overall, we find that models perform similarly on average for single-concept and dual-concept patterns, which indicates that they also struggle with puzzles that require reasoning about multiple abstract concepts.

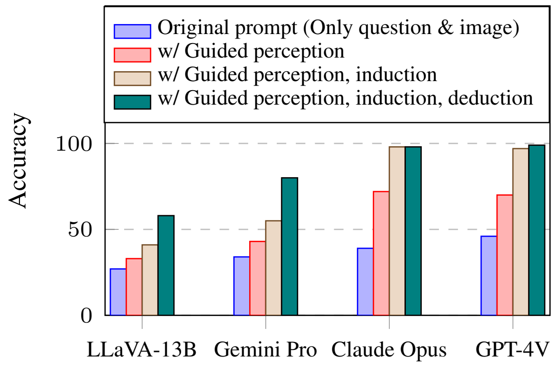

Figure 4: Analysis on multimodal reasoning bottlenecks. We progressively prompt models with ground-truth explanations for visual perception, inductive reasoning, and deductive reasoning.

5.1 Analysis of Multimodal Reasoning Bottlenecks

Given the lower performance of existing large multimodal models, this raises the natural question of why they are not able to reason well about abstract patterns. As shown in Figure 2, the stages of solving abstract puzzles can be generally decomposed into visual perception, inductive reasoning, and deductive reasoning. Hence, we analyze their reasoning bottlenecks in Figure 4 by progressively providing the ground truth explanation in their prompts. Note that we omit the final answer in the deductive reasoning explanation to avoid making the question trivial, and the detailed prompts can be found in the supplementary material. Overall, we observe that the models perform better when provided with ground truth explanations, which suggests that they are able to leverage the additional information.

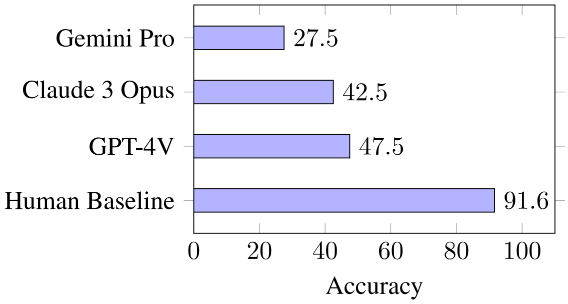

Figure 5: Comparison between average human performance and large multimodal models on a subset of PUZZLEVQA.

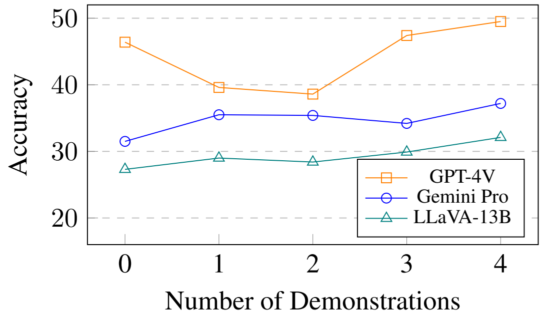

Figure 6: Analysis on the effect of few-shot demonstrations on model performance for single-concept puzzles.

Notably, GPT-4V and Claude 3 Opus is able to solve almost all cases when provided with both visual perception and inductive reasoning explanations. This suggests that the main bottlenecks for GPT-4V and Claude 3 Opus are visual perception and inductive reasoning to interpret the multimodal information and recognize the pattern from observations. However, this is not the case for LLaVA-13B and Gemini Pro, which demonstrate the largest improvement when guided by visual perception, inductive reasoning, and deductive reasoning together. This indicates that their main bottleneck is deductive reasoning to apply general principles of the pattern to solve specific cases. Note that these results are intended to serve as an **optimistic upper bound** of the model performance when provided with ground truth partial information, and may not indicate that the puzzles will become trivial.

5.2 Comparison to Human Performance

To further shed light on how the large multimodal models compare to the reasoning ability of humans, we conducted a human baseline study involving 23 university students[4]. Participants were allotted 30 minutes to solve 40 puzzle instances sampled from our 20 puzzle categories, yielding an average human baseline score of 91.6%, as shown in Figure 5. Note that the 20-24 age group of the participants correspond to the formal operational stage of cognitive development, as discussed in Section 2. In contrast, the highest-performing model, GPT-4V scored 47.5% on the same set of puzzle samples, highlighting the specific bottlenecks causing models to fall short of human cognition: primarily in visual perception and inductive reasoning, as discussed in Section 5.1.

[4]Note that the participants volunteered for the short study and we obtained prior permission from their instructor.

5.3 Effect of Few-Shot Demonstrations

While we focus on the zero-shot setting to investigate how multimodal models handle novel reasoning challenges, we also explore how models may use knowledge and strategies from other puzzles to solve a new, specific puzzle. This is akin to analogical reasoning, which involves using experience from similar scenarios to make inferences about a novel situation (Webb et al., 2022). Concretely, we run an analysis to study the effect of in-context learning (Brown et al., 2020) with few-shot demonstrations of other puzzle instances when the model is tasked to solve a specific puzzle. The demonstrations consist of interleaved instances of multimodal inputs of each puzzle image and question, as well as the ground truth reasoning explanations. To ensure that the demonstrations are sufficiently diverse, we randomly select puzzles of different categories from the given puzzle. As shown in Figure 6, we find a general trend of increasing performance with respect to the number of demonstrations. Although there are some cases of lower performance for GPT-4V, we see that models generally achieve their best performance with the most number of demonstrations. This suggests that the models are indeed capable of analogical reasoning, and in-context learning may be a promising direction to enhance the abstract reasoning abilities of multimodal models in the future.

In addition, while not the main focus of this work, there may be other methods of improving model performance, including model training or different prompting methods. Hence, we have also included preliminary studies on fine-tuning with LLaVA-13B and comparison between chain-of-thought and direct prompting in the supplementary material. To consider the effect of other factors, the supplementary material further includes an alternative setting with text-only models given the ground-truth visual perception caption, and an analysis on the effect of evaluation dataset size.

6 Related Work

The recent surge in multimodal pretraining and fine-tuning approaches (Liu et al., 2023; Bai et al., 2024) has led to the creation of various benchmarks. Benchmarks like VQA (Antol et al., 2015) and OK-VQA (Marino et al., 2019) aim at evaluating the basic perception and reasoning abilities of large multimodal models. Meanwhile, benchmarks like MMMU (Yue et al., 2023) and ScienceQA (Lu et al., 2022) offer an evaluation of LLMs' proficiency across multiple disciplines requiring domain-specific knowledge and multimodal understanding.

To investigate the fundamental challenges in multimodal perception and reasoning, we deliberately focus on the abstract domain, aiming to assess how models emulate cognitive abilities, particularly involving reasoning about abstract concepts and relationships. However, we note that existing benchmarks have limitations which make them less suitable for studying large multimodal models. The RAVEN dataset (Zhang et al., 2019) presents visual matrices with abstract patterns, challenging models to identify patterns and complete missing elements. However, we note that it has a specific spatial layout and can be solved exactly with search algorithms. Compared to CLEVR (Johnson et al., 2017) which offers synthetic visual scenarios and questions focusing on logic and commonsense, we focus on exploring how large multimodal models perceive and reason about multimodal patterns, which is more closely related to fundamental cognitive processes in humans (Mattson, 2014). While ConceptARC (Moskvichev et al., 2023) focuses on specific spatial concepts such as inside-outside and above-below, our dataset PUZZLEVQA studies how visual objects interact and relate based on broader abstract concepts such as colors, shapes, numbers, and size. Lastly, the MiniSCAN dataset (Lake et al., 2019) presents patterns that map special words to a sequence of color symbols, but is limited to color-based patterns.

What distinguishes our dataset, PUZZLEVQA, from the existing works is its systematic analysis of multimodal reasoning through abstract patterns, including perceptual, inductive, and deductive reasoning. Compared to the previous datasets, our multimodal patterns encompass broad and fundamental abstract concepts such as numbers, colors, shapes, and size. Notably, our dataset not only provides ground truth answers but also includes image captions and pattern explanations that enable more detailed and systematic diagnosis of the reasoning bottlenecks for large multimodal models.

7 Conclusion

In this work, we introduced the PUZZLEVQA dataset to investigate the reasoning challenges in large multimodal models. Our experiments demonstrated that, despite their sophistication, models such as GPT-4V exhibit substantial challenges when solving abstract pattern puzzles that require visual perception, inductive reasoning, and deductive reasoning, falling short of cognitive processes displayed by humans. Notably, our systematic analysis with ground truth explanations reveals that the main reasoning bottlenecks for GPT-4V are weaker visual perception and inductive reasoning capabilities. On the other hand, we found that other large multimodal models required more guidance with ground truth explanations, pointing to a broader range of reasoning challenges. Looking ahead, our work points to exciting avenues for advancing the reasoning abilities of large multimodal models. Future research should focus on enhancing models' understanding of multimodal information and refining their abstract reasoning faculties, in order to further enhance their general capabilities.

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